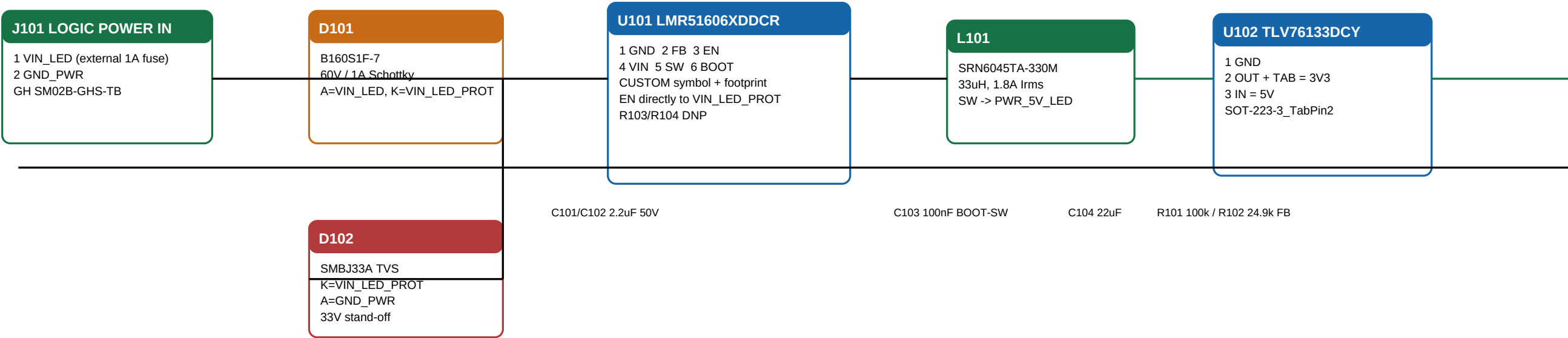


HUMAN REVIEW ONLY - source: LED_CAN_NODE_KICAD_ENTRY_REFERENCE.md - not a fabrication release



[FIXED] TAPE POWER IS EXTERNAL

Tape +24V and power GND: distribution -> external 5A fuse -> tape input.
They do not enter this PCB. J101 powers node electronics only.

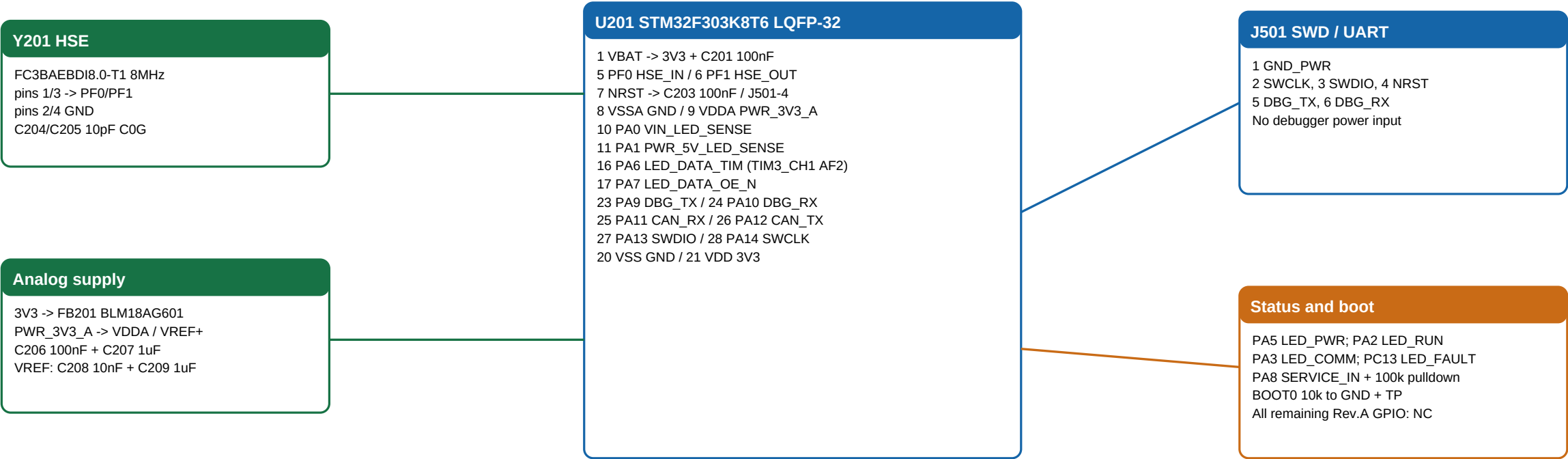
[HUMAN CHECK] BUCK REVIEW

BOOT capacitor must be BOOT-to-SW. Verify custom U101 pin order and pad-1.

[UNRESOLVED] CURRENT LIMIT

5A fuse is a protective harness maximum, not a declared tape current.
Measure all-white current before changing fuse or firmware brightness limit.

HUMAN REVIEW ONLY - source: LED_CAN_NODE_KICAD_ENTRY_REFERENCE.md - not a fabrication release



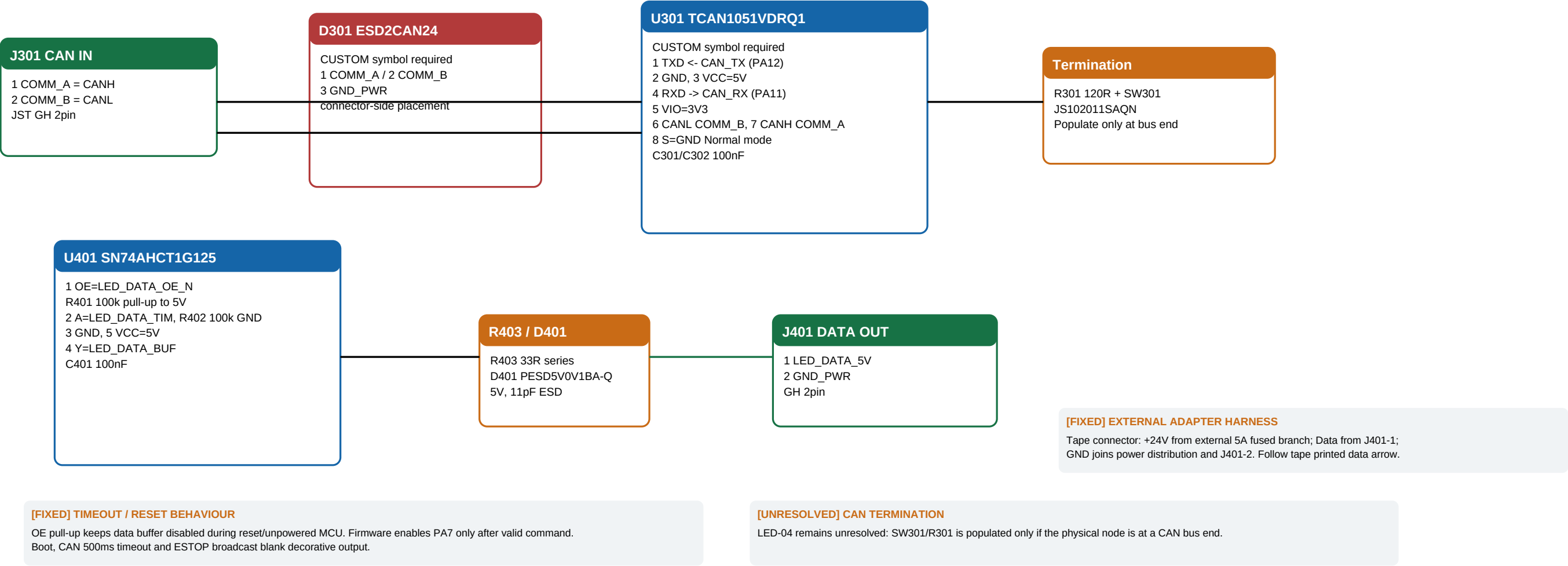
[FIXED] POWER PINS

VDD/VBAT=3V3. VSS/VSSA=GND. Every VDD has 100nF local decoupling.
Do not infer unused pin connections: follow the entry-table NC rows.

[HUMAN CHECK] HUMAN PIN REVIEW

Compare every LQFP-32 pin number in this sheet to DS9866 before KiCad entry.
ERC cannot prove a hand-transcribed package pin number.

HUMAN REVIEW ONLY - source: LED_CAN_NODE_KICAD_ENTRY_REFERENCE.md - not a fabrication release



External harness - tape power

24V distribution
-> external 5A automotive fuse
-> Tape +24V / Tape power GND
2m, 630 packages/m, 24V WS2811
Single injection at data-input end
Manual says add injection every 5m

External harness - node

24V branch -> external 1A fuse -> J101
J301: COMM_A/COMM_B only
J401-1: 5V buffered data
J401-2: GND data reference
Adapter merges J401 with supplied 3pin tape plug

Control boundary

Central sends high-level CAN effects
F303 timer+DMA creates WS2811 waveform
No continuous RGB frames on CAN
Node ID 6 / Classic CAN 1Mbps
This tape is not safety indication

ERC-non-detectable human review

- ☐ U101 DBV pin order and pad-1: BOOT capacitor is BOOT-to-SW.
- ☐ U102 tab/pin2 is 3V3; pin3 is 5V.
- ☐ TCAN direction: PA12->TXD1, RXD4->PA11; VCC=5V, VIO=3V3, S=GND.
- ☐ AHCT OE is active-low and pulled to 5V; reset cannot enable data.
- ☐ J401 carries only buffered 5V Data and GND reference - never 24V.
- ☐ Tape data arrow points away from controller; 24V/power GND bypass PCB.
- ☐ Termination fitted only at the actual physical end of the CAN bus.

[UNRESOLVED] RELEASE BLOCKERS

LED-01: measure tape all-white current and pixel-group count.
LED-03: EN UVLO divider stays DNP; direct EN population is approved.
LED-04: confirm bus-end location before fitting termination.

[FIXED] NEXT MECHANICAL STEPS

1. Enter one zone at a time from the controlled Markdown table.
2. ERC after each zone. 3. Export netlist and diff against pin table.
4. Complete the checklist above. 5. Only then begin PCB placement.